

Commerce XI Maths App

Exercise 1.3

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 18

Question 1 - Q1(i)

The following table gives frequency distribution of marks of 100 students in an examination.

Marks	15-20	20-25	25-30	30-35	35-40	40-45	45-50
No. of students	9	12	23	31	10	8	7

Determine Q_1 graphically.

Correct answer: 26

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 26.

Question 2 - Q1(ii)

The following table gives frequency distribution of marks of 100 students in an examination.

Marks	15-20	20-25	25-30	30-35	35-40	40-45	45-50
No. of students	9	12	23	31	10	8	7

Determine D_6 graphically.

Correct answer: 32.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 32.5.

Question 3 - Q1(iii)

The following table gives frequency distribution of marks of 100 students in an examination.

Marks	15-20	20-25	25-30	30-35	35-40	40-45	45-50
No. of students	9	12	23	31	10	8	7

Determine P_{85} graphically.

Correct answer: 40

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 40.

Question 4 - Q2

The following table gives the distribution of daily wages of 500 families in a certain city.

Daily wages	Below 100	100–200	200–300	300–400	400–500	500–600	600 and above
No. of families	50	150	180	50	40	20	10

Draw a 'less than' ogive for the above data. Determine the median income and obtain the limits of income of central 50% of the families.

Options

A. 150, 228, 298

B. 150, 228, 300

C. 148, 225, 298

D. 150, 250, 350

Correct answer: 150, 228, 298

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 150, 228, 298.

Question 5 - Q3

From the following distribution, determine median graphically.

Daily wages (in ₹)	Above 300	Above 400	Above 500	Above 600	Above 700	Above 800	Above 900
No. of employees	520	470	399	210	105	45	7

Correct answer: 574

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 574.

Question 6 - Q4(i)

The following frequency distribution shows the profit (in ₹) of shops in a particular area of city.

Profit per shop (in ₹000)	0-10	10-20	20-30	30-40	40-50	50-60
No. of shops	12	18	27	20	17	6

Find graphically the limits of middle 40% shops:

Options

A. 20, 36.5

B. 20, 35

C. 18, 36.5

D. 22, 38

Correct answer: 20, 36.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 20, 36.5.

Question 7 - Q4(ii)

The following frequency distribution shows the profit (in ₹) of shops in a particular area of city.

Profit per shop (in ₹000)	0-10	10-20	20-30	30-40	40-50	50-60
No. of shops	12	18	27	20	17	6

Find graphically the number of shops having profit less than ₹35,000.

Correct answer: 67

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 67.

Question 8 - Q5(i)

The following is frequency distribution of over time (per week) performed by various workers from a certain company.

Overtime (in hours)	Below 8	8-12	12-16	16-20	20-24	24 and above
No. of workers	4	8	16	18	20	14

Determine D_2 graphically.

Correct answer: 13

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 13.

Question 9 - Q5(ii)

The following is frequency distribution of over time (per week) performed by various workers from a certain company.

Overtime (in hours)	Below 8	8-12	12-16	16-20	20-24	24 and above
No. of workers	4	8	16	18	20	14

Determine Q_2 graphically.

Correct answer: 19

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 19.

Question 10 - Q5(iii)

The following is frequency distribution of over time (per week) performed by various workers from a certain company.

Overtime (in hours)	Below 8	8-12	12-16	16-20	20-24	24 and above
No. of workers	4	8	16	18	20	14

Determine P_{61} graphically.

Correct answer: 21.2

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 21.2.

Question 11 - Q6(i)

Draw ogive for the following data and hence find the value of D_1 .

Marks less than	10	20	30	40	50	60	70	80	90
No. of students	4	6	24	46	67	86	96	99	100

Correct answer: 22

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 22.

Question 12 - Q6(ii)

Draw ogive for the following data and hence find the value of Q_1 .

Marks less than	10	20	30	40	50	60	70	80	90
No. of students	4	6	24	46	67	86	96	99	100

Correct answer: 30.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 30.5.

Question 13 - Q6(iii)

Draw ogive for the following data and hence find the value of P_{40} .

Marks less than	10	20	30	40	50	60	70	80	90
No. of students	4	6	24	46	67	86	96	99	100

Correct answer: 37

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 37.

Question 14 - Q7

The following table shows the age distribution of head of the families in a certain country.

Age of head of family (in years)	Under 35	35-45	45-55	55-65	65-75	75 and above
Numbers (million)	46	85	64	75	90	40

Determine the third, fifth and eighth decile of the distribution graphically.

Options

A. 44, 55.5, 68

B. 42, 55.5, 70

C. 44, 57.5, 68

D. 45, 56, 70

Correct answer: 44, 55.5, 68

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 44, 55.5, 68.

Question 15 - Q8

The following table gives the distribution of females in an Indian village.

Age group	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
No. of females (in 000)	175	100	68	48	25	50	23	8	2	1

Determine the median age graphically.

Correct answer: 17.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 17.5.

Question 16 - Q9

Draw ogive for the following distribution and hence find graphically the limits of weight of middle 50% fishes.

Weight of fishes (in gms)	800-890	900-990	1000-1090	1100-1190	1200-1290	1300-1390	1400-1490
No. of fishes	8	16	20	25	40	6	5

Options

A. 1025, 1248

B. 1000, 1250

C. 1025, 1250

D. 1050, 1248

Correct answer: 1025, 1248

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 1025, 1248.

Question 17 - Q10(i)

Find graphically the values of D_3 for the data given below :

I.Q. of students	60-69	70-79	80-89	90-99	100-109	110-119	120-129
No. of students	20	40	50	50	20	10	10

Correct answer: 79.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 79.5.

Question 18 - Q10(ii)

Find graphically the values of P_{65} for the data given below :

I.Q. of students	60-69	70-79	80-89	90-99	100-109	110-119	120-129
No. of students	20	40	50	50	20	10	10

Correct answer: 93.5

Explanation and steps

Form the appropriate cumulative frequencies, draw the required ogive and read the partition value from the graph. The textbook value is 93.5.